

CSE4356/CSE5392/EE5315/EE4328 (Fall 2025)

Lab #5

In this lab, you will write Verilog code to implement the I2C serial interface and the receive FIFO.

The following steps will guide you through this process:

- 1.** Implement the CONTROL register, outputting signals to control the FSM.
- 2.** Output the test output if enabled with TEST_OUT.
- 3.** Leave the transmit FIFO instance from Lab 4 with the write port connected to the DATA register and status bits routed to the STATUS register. Add logic so that writing a '1' to the STATUS register TXFO bit clears an overflow condition.
- 4.** Add a receive FIFO instance with the read port connected to the DATA register and status bits routed to the STATUS register. Add logic so that writing a '1' to the STATUS register RXFO bit clears an overflow condition.
- 5.** Design an I²C serial interface that send a clock on the I2C_SCL pin and transmits and receives data on the I2C_SDA pin. I2C_SCL and I2C_SDA should be an open drain interfaces. The serial interface should idle with both pins floating (they will be high due to the external pull-up resistors).
- 6.** Connect the read port of the transmit FIFO and the write port of the receive FIFO to a I²C serial interface. Connect the START, USE_REPEATED_START, USE_REGISTER from the CONTROL register to the interface. Connect the ACK_ERROR and BUSY bits in the status register to the I²C serial interface. Also provide the ability to write to the ACK_ERROR bit in the status register to clear an error condition. When the FIFO is in the idle state and the START bit is set, it should leave the idle state, send a START condition, and start the bus transaction following the steps in the main project PDF. After BYTE_COUNT bytes of data are sent or received, then the STOP condition is sent and idle state is re-entered.
- 7.** Connect the TEST_OUT to a scope the correct frequency is being generated.
- 8.** Connect SDA and SCL to the MCP23008 IC. Pullups should be connected to both outputs. At least one output and one input must be connected the the MCP23008 to show operation. Write a program similar to the gpio utility and stop go program to test operation.
- 9.** Please implement all features of the project and submit by the due date in the syllabus.
- 10.** Send the i2c.v file and C programs to the TA.
- 11.** This is last interim milestone for the project. You should review the project to verify that all other features, other than sysfs are implemented.

This lab is due by November 23, with a 20% penalty per week day for being late.